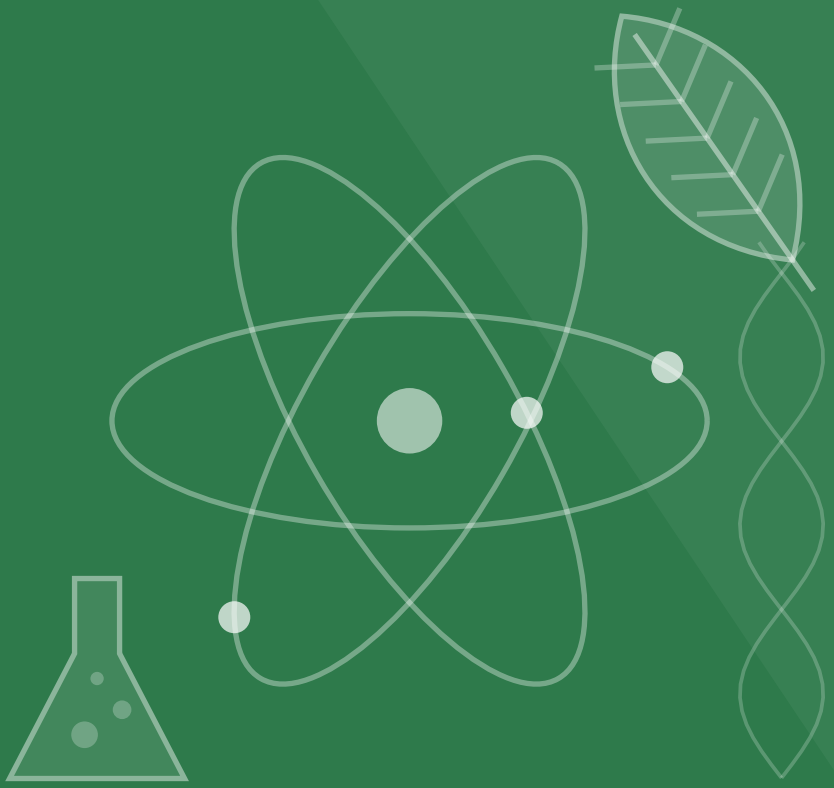
VENVLEARNING

STUDENT WORKBOOK



Science

GRADE 7

Daily practice, weekly checks, and unit tests

2026–2027 School Year · Unit 1: Cycles of Matter and Energy

VENVLEARNING

Grade 7
Science

STUDENT WORKBOOK

2026–2027 School Year

Unit 1 Cycles of Matter and Energy

This book belongs to _____

A VenvLearning family homeschool curriculum

How to use this workbook

Every lesson in your textbook has matching practice pages here. Show your thinking in the work space. When a problem has an answer line, write your final answer there.

- Warm-Up reviews something you already know.
- Practice uses today's lesson.
- Think Deeper problems ask you to explain or find a mistake.
- Exit Ticket shows your teacher what you learned today.

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UNIT 1

Cycles of Matter and Energy

Science practices, Earth's spheres, the water, carbon, oxygen, and nitrogen cycles, photosynthesis, respiration, food webs, and biodiversity.

Your practice pages for Unit 1. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

independent variable	The variable that is changed.
dependent variable	The variable that is measured.
photosynthesis	Light + CO_2 + water → glucose + oxygen.
cellular respiration	Glucose + oxygen → CO_2 + water + energy.
nitrogen fixation	Changing N_2 into usable forms.
biodiversity	The variety of life in an area.

UNIT 1 · LESSON 1.1 · PRACTICE

Lesson 1.1: Variables, Data, and Scientific Explanations

Name _____ Date _____

I can identify independent, dependent, and controlled variables and write a scientific explanation using CER.

Warm-Up

1. The variable you change

2. The variable you measure

Identify the Variables

3. Does the height of a ramp affect how far a car rolls? Independent:

4. Same: dependent:

5. Same: two controlled variables:

6. Write a hypothesis for the ramp experiment.

Analyze Data

7. Data: 0 g salt → water boils at 100.0°C; 20 g → 100.3°C; 40 g → 100.7°C. Write a CER.

Exit Ticket

8. Why must controlled variables stay the same?

UNIT 1 · LESSON 1.2 · PRACTICE

Lesson 1.2: Earth's Spheres and the Water Cycle

Name _____ Date _____

I can describe Earth's four spheres and model how energy from the sun and gravity drive the water cycle.

Warm-Up

1. All living things make up the...

2. Water vapor cooling into droplets is...

Practice

3. Which sphere does a river belong to?

4. Which sphere does a mountain belong to?

5. What provides the energy for evaporation?

6. What happens to water that soaks into soil?

7. Describe one interaction between two spheres in the water cycle.

8. Explain the droplets on the inside of the bag using the words evaporation and condensation.

Exit Ticket

9 Name the two things that drive the water cycle and what each does.

WEEKLY CHECK · Fri Oct 16 · ABOUT 15 MINUTES

Weekly Check 1: Variables and the Water Cycle

Name _____ Date _____

Show what you know

1. Does the amount of sunlight affect bean plant growth? Independent variable:

2. Dependent variable:

3. One controlled variable:

4. Water vapor cooling into droplets is...

5. Which sphere includes all living things?

6. Name the two drivers of the water cycle and what each does.

UNIT 1 · LESSON 1.3 · PRACTICE

Lesson 1.3: The Carbon and Oxygen Cycles

Name _____ Date _____

I can explain how carbon and oxygen cycle among living things, the atmosphere, the oceans, and rocks.

Warm-Up

1. Coal, oil, and natural gas are...

2. Organisms that break down dead matter

Practice

3. A tree takes in CO_2. Process?

4. A car burns gasoline. Process?

5. A fungus breaks down a log. Process?

6. You exhale. Process?

7. Explain why burning fossil fuels adds carbon to the atmosphere faster than natural processes remove it.

Exit Ticket

8. How are the carbon and oxygen cycles connected?

UNIT 1 · LESSON 1.4 · PRACTICE

Lesson 1.4: The Nitrogen Cycle

Name _____ Date _____

I can explain how nitrogen moves from the air into living things and back.

Warm-Up

1 Living things use nitrogen to build...

2 A bean plant is a...

Practice

3 Can animals use nitrogen gas directly?

4 How do animals get nitrogen?

5 Name a non-living way nitrogen is fixed.

6 What returns nitrogen from dead organisms to soil?

7 Explain why rotating crops with legumes can help soil.

Exit Ticket

8 Trace one nitrogen atom from the air into your body.

WEEKLY CHECK · Fri Oct 23 · ABOUT 15 MINUTES

Weekly Check 2: Carbon, Oxygen, and Nitrogen

Name _____ Date _____

Show what you know

1. Which process removes CO_2 from air?

2. Name a process that releases CO_2.

3. About what percent of air is nitrogen?

4. What do nitrogen-fixing bacteria do?

5. Why does burning fossil fuels change the carbon cycle?

UNIT 1 · LESSON 1.5 · PRACTICE

Lesson 1.5: Photosynthesis

Name _____ Date _____

I can explain how plants use light energy to make sugar and oxygen from carbon dioxide and water.

Warm-Up

1. Reactants of photosynthesis

2. Organelle where it happens

Analyze Data

3. Disks floating after 10 min: lamp cup 9; dark cup 0. Which had more photosynthesis?

4. What gas made the disks float?

5. What was the independent variable?

6. Write a CER for the leaf disk results.

Exit Ticket

7. Explain how photosynthesis transforms energy.

UNIT 1 · LESSON 1.6 · PRACTICE

Lesson 1.6: Cellular Respiration

Name _____ Date _____

I can explain how cells release energy from food through cellular respiration.

Warm-Up

1 Products of respiration

2 Reactants of respiration

Compare

3 Which process stores energy in glucose?

4 Which process releases energy from glucose?

5 Which produces oxygen?

6 Which produces carbon dioxide?

7 Balloon A (sugar) grew to 28 cm; balloon B (no sugar) to 13 cm. Explain using CER.

Exit Ticket

8 Explain how matter cycles but energy flows between photosynthesis and respiration.

WEEKLY CHECK · Fri Oct 30 · ABOUT 15 MINUTES

Weekly Check 3: Photosynthesis and Respiration

Name _____ Date _____

Show what you know

1. Products of photosynthesis

2. Products of cellular respiration

3. Organelle for photosynthesis

4. Organelle for respiration

5. Explain how photosynthesis and respiration cycle matter.

UNIT 1 · LESSON 1.7 · PRACTICE

Lesson 1.7: Food Webs and Energy Pyramids

Name _____ Date _____

I can model how energy flows through a food web and explain why less energy reaches each higher level.

Warm-Up

1 An organism that makes its own food

2 Organisms that break down dead matter

Practice

3 Producers have 50,000 units. Primary consumers?

4 Secondary consumers?

5 Tertiary consumers?

6 In grass → rabbit → fox, the fox is a…

7 If a disease kills most rabbits, predict effects on the grass and the foxes. Explain.

Exit Ticket

8 Explain the difference between how matter and energy move through an ecosystem.

UNIT 1 · LESSON 1.8 · PRACTICE

Lesson 1.8: Biodiversity and Ecosystem Services

Name _____ Date _____

I can explain why biodiversity matters and evaluate ways to protect ecosystem services.

Warm-Up

1. An animal that carries pollen

2. Wetlands help prevent…

Practice

3. Lawn square: 3 species. Field square: 14 species. Which has more biodiversity?

4. Explain one way the loss of pollinators would affect people.

5. Evaluate: A farmer wants to remove all hedgerows to plant more crops. Name one benefit and one cost to ecosystem services.

Exit Ticket

6. Choose one ecosystem service and explain how people could protect it.

UNIT REVIEW · Thu Nov 5 (homework) or as needed · ABOUT 40 MINUTES

Unit 1 Review

Name _____ Date _____

Practices and cycles

1. The variable you measure

2. Plants releasing water vapor

3. Changing N_2 into usable forms

4. Burning fuel

Energy

5. Photosynthesis equation (words)

6. Producers 30,000 units; primary consumers?

7. Why do energy pyramids narrow at the top?

UNIT TEST · Fri Nov 6 · ABOUT 45 MINUTES

Unit 1 Test

Name _____ Date _____

Show your work. This test has 20 points.

Part A · Science practices (1 point each)

1. Does the temperature of water affect how much sugar dissolves? Independent variable:

2. Dependent variable:

3. Name one controlled variable.

Part B · Cycles (1 point each)

4. Which process adds water vapor to the air from plants?

   A condensation
   B transpiration
   C infiltration
   D runoff

5. What force pulls precipitation down?

6. Which process removes carbon dioxide from the atmosphere?

   A combustion
   B decomposition
   C photosynthesis
   D respiration

7. Why can't animals use nitrogen gas directly?

8. Name the organisms that return nutrients from dead matter to soil.

Part C · Energy in living things (1 point each)

9 Write the word equation for photosynthesis.

10 Write the word equation for cellular respiration.

11 Producers store 40,000 units of energy. About how much reaches secondary consumers?

12 Which is an ecosystem service?

A a parking lot
B wetlands filtering water
C a concrete channel

Part D · Data analysis (4 points)

13 Leaf disks floating after 12 minutes: bright light 10, dim light 5, dark 0. Write a CER explaining the results.

Part E · Explain (4 points)

14 Use a food chain (grass → rabbit → fox) to explain how matter cycles and energy flows. Include photosynthesis, respiration, and decomposers.

Grade 7 Science

Student Workbook · 2026–2027

A full year of Grade 7 science built around questions, investigations, and evidence. Students observe, measure, model, and explain the natural world, then use what they learn to design solutions to real problems.

This Student Workbook holds the daily practice pages for every lesson, plus each Friday's Weekly Check, the unit reviews, and the unit tests, with room to show thinking and write complete answers.

INSIDE THIS YEAR

01 Cycles of Matter and Energy

IN THIS BOOK

- Warm-ups, practice, and exit tickets for every lesson
- Weekly Checks every Friday
- Unit reviews and unit tests
- Space to show work and explain thinking

THE COMPLETE SET



Textbook

Workbook

You are holding this book

Teacher Edition

Each lesson has the same number in all three books, and every day of the year is mapped to its lesson in the VenvLearning planner.